



Pamantasan ng Lungsod ng Maynila



AN INTERACTIVE WASTE SEGREGATION LEARNING PROTOTYPE FOR
CHILDREN USING MOBILENETV2-BASED MULTI-FRAME IMAGE
CLASSIFICATION

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By
Jan Leander G. Jacobe
Laurence King S. Calvan
Charles Hendricks D. Solinap

Engr. Eufemia A. Garcia
Research Adviser

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Solid waste management remains a critical environmental and public health concern in many urban communities, particularly in developing nations where rapid population growth, increased consumption of packaged products, and inadequate infrastructure converge to produce growing volumes of improperly segregated municipal solid waste (Shreya et al., 2023; Dokl et al., 2025). In the Philippines, the problem is further compounded by low compliance with Republic Act No. 9003 (2001), also known as the Ecological Solid Waste Management Act, which mandates the segregation of solid waste at the source into biodegradable, recyclable, and residual categories. Despite the existence of this legal framework, actual segregation behavior among households and institutions remains inconsistent, with contaminated waste streams presenting significant challenges for recycling facilities and local government waste processors (Republic Act No. 9003, 2001; Arishi, 2025).

To address the technical limitations of waste identification, researchers and engineers have explored image-based classification systems that employ computer vision and deep learning to automate the categorization of waste materials. Convolutional neural networks (CNNs), particularly lightweight architectures such as MobileNetV2, have demonstrated strong performance in waste recognition tasks while remaining computationally efficient enough for real-time deployment on edge devices (Sandler et al., 2018; Nzeuga et al., 2025). Furthermore, multi-frame prediction aggregation has been proposed as a method to improve classification stability in the presence of visual noise, contamination, and varying lighting conditions inherent in real-world waste disposal environments (Yang et al., 2023; Nguyen et al., 2024).



While these technical advances are promising, a critical and underexplored gap remains: the behavioral and educational dimension of waste segregation. Automated classification systems address the efficiency of waste sorting after waste is generated, but they do not address the root cause of improper segregation, namely, the absence of a deeply ingrained habit of correct waste disposal from an early age. Research in environmental education consistently identifies early childhood as the most formative period for establishing sustainable behavioral patterns, including waste segregation practices (Barraza, 1999; Cheng & Monroe, 2012). Children who are exposed to correct segregation principles through interactive, hands-on experiences during their elementary years are significantly more likely to internalize and practice these behaviors into adulthood and within their households (Kollmuss & Agyeman, 2002; Ernst & Monroe, 2004).

Existing educational materials for waste segregation in Philippine elementary schools, however, rely predominantly on traditional lecture-based instruction and printed visual aids. These passive methods have limited efficacy in producing lasting behavioral change because they do not engage the sensorimotor and experiential learning pathways that are most active during early childhood development (Kolb, 1984; Dewey, 1938). Interactive, physical prototypes that provide immediate, automated feedback — where a child places an object and the device responds with a clear classification — represent a category of experiential learning tool that has demonstrated superior retention and engagement in comparable environmental education contexts (Bers et al., 2014; Resnick & Rosenbaum, 2013). The integration of an AI-powered image classification system into such a physical device creates an opportunity to bridge the gap between advanced waste recognition technology and early childhood environmental education.

This study, therefore, proposes the design and implementation of a physical, interactive educational prototype for elementary-aged children that employs MobileNetV2-based multi-frame image classification to automatically identify and display the correct segregation category of a presented waste object in accordance with the biodegradable, recyclable, and residual categories of RA 9003. By deploying the AI



classification system on a tangible, child-accessible hardware platform, this research aims not only to validate the technical performance of the classification pipeline but also to demonstrate its effectiveness as a practical educational intervention for children in community and school-based settings.

1.2 Statement of the Problem

Effective solid waste management in the Philippines is critically undermined by persistent non-compliance with the source segregation mandate of Republic Act No. 9003 (2001). Despite the legal requirement to separate waste into biodegradable, recyclable, and residual categories, actual segregation behavior among households and institutions remains inconsistent, with contaminated waste streams continuing to degrade recycling efficiency and increase the operational burden on local waste processors (Arishi, 2025; Dokl et al., 2025).

A primary technical obstacle to automated waste segregation support is the unreliability of existing image-based classification systems under real-world disposal conditions. Most systems are trained and evaluated on clean benchmark datasets that do not reflect the visual degradation caused by food residue, liquid contamination, variable lighting, and irregular object geometry encountered in actual disposal environments (Miski et al., 2023; Zhao et al., 2022; Pucnik et al., 2024). Compounding this, the majority of existing systems rely on single-frame predictions, which produce unstable classification outputs when waste objects are captured under conditions of partial occlusion, motion blur, or lighting variation (Dipo et al., 2025; Ling et al., 2026). These limitations reduce the practical reliability of automated waste recognition in the uncontrolled settings where it is most needed.

Beyond the technical dimension, a deeper and less-addressed gap exists in the behavioral formation of waste segregation habits. Current environmental education approaches in Philippine elementary schools rely predominantly on lecture-based instruction and static printed materials — passive modalities that have been shown to be insufficient for establishing durable, procedural waste segregation behavior in young



learners (Cheng & Monroe, 2012; Rickinson et al., 2004). Early childhood has been consistently identified as the most critical window for environmental habit formation, yet no locally developed, hardware-based interactive tool currently exists that bridges AI-powered waste classification with experiential, child-centered learning aligned with RA 9003 categories (Barraza, 1999; Kollmuss & Agyeman, 2002).

This study addresses these gaps through the following research questions:

1. How to develop a waste classification system that can accurately classify waste into biodegradable, recyclable, residual and others?
2. How effective is the proposed system in segregating waste into the biodegradable, recyclable, residual, and others?
3. How can waste classification systems be integrated into an interactive prototype to serve as an educational tool for elementary-aged children learning waste segregation?

1.3 Objectives of the Study

1.3.1 General Objectives

This study aims to develop and evaluate a MobileNetV2-based multi-frame image classification system for contaminated solid waste, and to investigate its integration into a physical interactive prototype as an educational tool for elementary-aged children aligned with the waste segregation categories of Republic Act No. 9003.

1.3.2 Specific Objectives

To achieve the general objective, this study aims to accomplish the following specific objectives:

1. Develop an image-based waste classification system capable of classifying contaminated solid waste images into biodegradable, recyclable, residual, and others (unrecognizable) categories from captured image frames using the MobileNetV2 convolutional neural network.



2. To design and construct a physical interactive prototype that integrates the image-based waste classification system with appropriate input and output hardware components to enable real-time, automated waste category feedback to a child user.
3. To evaluate the educational utility of the prototype by assessing the correctness of waste category identification in observed interactions with elementary-aged children in a controlled demonstration context.

1.4 Significance of the Study

This study on the development of a multi-frame image-based classification system for contaminated solid waste using MobileNetV2 provides valuable contributions to various sectors, including industry, academe, government, researchers, and special interest groups. By improving the reliability of automated waste classification systems, the study supports efforts toward more efficient waste management and environmental sustainability.

Industry.

For industries involved in waste management, recycling, and environmental technology, the findings of this study may contribute to the development of more efficient automated waste sorting systems. By utilizing computer vision and deep learning techniques, the proposed classification system can help reduce manual labor in waste sorting processes while improving sorting accuracy (Pitakaso et al., 2024). The integration of multi-frame image analysis may also enhance the reliability of automated classification systems, particularly when dealing with contaminated solid waste. As a result, industries engaged in recycling operations may benefit from improved material recovery rates and more efficient waste processing workflows.



Academe.

In the academic community, this study contributes to the growing body of knowledge in the fields of computer vision, machine learning, and intelligent waste management systems. The research demonstrates how deep learning models such as MobileNetV2 can be applied to real-world environmental problems, particularly in the area of waste classification. Furthermore, the study introduces the concept of multi-frame prediction aggregation as a method for improving classification reliability under conditions of visual variability and contamination. The findings may serve as instructional material for students and researchers studying artificial intelligence, image processing, and environmental engineering applications.

Government.

Government agencies responsible for environmental management and waste regulation may benefit from the insights provided by this research. In the Philippines, Republic Act No. 9003 (2001) mandates the segregation of solid waste into biodegradable, recyclable, and residual categories as a fundamental requirement for sustainable waste management. The proposed automated classification approach may support government initiatives aimed at promoting compliance with this mandate and reducing landfill waste. By improving the accuracy of waste classification systems, the study may also contribute to the development of technology-assisted waste management solutions that align with national and local environmental sustainability programs.

Other Researchers.

This study may serve as a useful reference for researchers who intend to explore image-based waste classification, computer vision applications, or deep learning methods in environmental systems. The methodology used in this research, including the implementation of MobileNetV2 and the application of multi-frame prediction aggregation, may provide a foundation for future studies aiming to improve classification performance in complex real-world conditions. Researchers may also build upon the



dataset preparation strategies and system design presented in this study to develop more advanced waste recognition systems.

Special Groups.

Environmental advocacy groups, recycling organizations, and community-based environmental initiatives may also benefit from the outcomes of this research. The development of automated waste classification technologies can support community efforts to improve waste segregation practices and raise awareness about responsible waste disposal. By providing technological tools that assist in identifying and categorizing waste materials, the study contributes to broader environmental protection efforts and encourages communities to participate more actively in sustainable waste management practices.

Local Schools and Educators.

Elementary school teachers, science educators, and school administrators stand to benefit directly from this study as it provides a concrete, deployable model for technology-assisted environmental instruction. The prototype offers educators a supplementary tool that transcends the limitations of chalk-and-board demonstrations by giving students immediate, automated, and accurate feedback on waste categorization. In a Philippine classroom context where environmental science is mandated under the K to 12 curriculum, a locally developed interactive device that reinforces RA 9003 classification in a tangible and engaging manner represents a meaningful addition to the instructional toolkit available to teachers, particularly in public elementary schools with limited laboratory or technology resources.

The Community and Future Generations.

At the broadest level, this study holds significance for Filipino communities and for the long-term sustainability of local waste management systems. Children who develop correct waste segregation habits through early experiential learning become agents of behavioral change within their families and neighborhoods. By targeting the formative



years of elementary education, the prototype functions not merely as a classroom device but as an intervention in the intergenerational transmission of environmental practices. Communities in which children have been educated through such tools are better positioned to achieve higher rates of source segregation, reduce recyclable contamination, and contribute to the national goals of the Ecological Solid Waste Management Act. The prototype thus represents an investment not only in the children who interact with it directly, but in the environmental health of Philippine communities for generations to come.

1.5 Scope and Limitations

This study encompasses two integrated components: the development and evaluation of an AI-based waste image classification system, and the design and implementation of a physical, interactive educational prototype incorporating that system for use by elementary-aged children.

The classification system component covers the training and evaluation of a MobileNetV2 convolutional neural network using two-phase transfer learning on waste image data mapped to the three categories mandated by Republic Act No. 9003 (2001): biodegradable, recyclable, and residual. The dataset used in this study consists of publicly available waste image sources, specifically the Kaggle Garbage Classification dataset (Abla, 2023) as the primary training source, with supplementary reference to the TrashNet dataset (Thung & Yang, 2016) for real-world visual variability. The classification framework is limited to these three broad functional waste categories and does not extend to fine-grained material-level classification. Multi-frame prediction aggregation employing a five-frame majority voting scheme is applied during real-time inference to improve classification stability. Model training, evaluation, and prototype demonstration are conducted in a personal computer-based environment using TensorFlow/Keras and OpenCV, with camera input delivered via the Iriun Webcam application streaming from a smartphone over a local Wi-Fi connection as a virtual webcam device.



The physical prototype component encompasses the hardware integration of the trained classification pipeline into an interactive, child-accessible device. The prototype is designed to present a classification result to the user through clear visual output upon presenting a physical waste object to the camera. The scope of the prototype evaluation is limited to demonstrating correct functionality and observing interactions with elementary-aged children (Grades 1 to 6, ages 6 to 12) in a controlled demonstration setting. The study does not conduct a formal longitudinal assessment of learning outcomes or behavioral change among child participants, as this falls within the scope of educational psychology research rather than electronics engineering. The prototype is not intended for gamified play; it is designed as a practical, instructional device that simulates a real-world waste bin interaction.

Several limitations must be acknowledged. The publicly available datasets used for model training may not fully represent the visual diversity of waste items encountered in Philippine household or school disposal conditions, particularly with respect to local packaging materials and contamination patterns. The hardware prototype is implemented at a development-level scale and does not constitute a fully ruggedized or mass-producible consumer device. Evaluation of the prototype's educational impact is observational and formative in nature, and does not constitute a controlled pedagogical experiment. Finally, the child demographic evaluation is limited to the elementary school age range, and findings regarding prototype effectiveness are not generalized to other age groups or learning contexts.

1.6 Definition of Terms

Biodegradable Waste. Refers to organic materials that can be naturally decomposed by microorganisms, including food scraps, yard waste, and other putrescible materials. In this study, biodegradable waste is one of the three waste categories used for image classification, in accordance with Republic Act No. 9003 (2001).

Computer Vision. This to the field of artificial intelligence that enables computers to interpret and analyze visual information from digital images or videos. In this study,



computer vision techniques are used to process images of waste objects for automated classification.

Contaminated Solid Waste. Refers to solid waste items whose surfaces have been soiled or altered by the presence of food residue, liquid waste, or other foreign materials, resulting in changes to their visual appearance that may complicate automated classification. In this study, contaminated solid waste represents the primary subject of classification, as the proposed system is specifically designed to handle waste images that exhibit surface contamination under real-world disposal conditions.

Convolutional Neural Network (CNN). A type of deep learning model designed to process image data by automatically learning hierarchical visual features. In this study, CNN architecture is used as the underlying mechanism for recognizing patterns in waste images.

Dataset. A structured collection of labeled images or data used to train and evaluate machine learning models. In this research, the dataset consists of publicly available waste images categorized into biodegradable, recyclable, and residual waste classes.

Deep Learning. A subset of machine learning that uses multi-layer neural networks to learn complex patterns from large datasets. In this study, deep learning techniques are applied to train the classification model to recognize different types of waste.

Educational Prototype. A functional, physical device designed for use as an instructional tool in a learning environment. In this study, the educational prototype refers to the hardware device that integrates the MobileNetV2 classification system and is designed to provide interactive waste category feedback to elementary-aged children.

Experiential Learning. A theory of learning, as formulated by Kolb (1984), that holds that knowledge is best acquired through direct, hands-on experience and reflection on that experience. In this study, the interactive prototype is designed to facilitate experiential learning by allowing children to receive immediate feedback on waste categorization decisions through physical interaction with the device.



Feature Extraction. The process of identifying and capturing important visual characteristics such as edges, shapes, and textures from images. In this study, the MobileNetV2 model performs feature extraction to identify distinguishing characteristics of waste objects.

Human-Computer Interaction (HCI). An interdisciplinary field concerned with the design, evaluation, and implementation of interactive computing systems for human use, with particular attention to usability, safety, and accessibility. In this study, HCI principles inform the design of the physical prototype to ensure that it is intuitive, safe, and appropriate for use by elementary-aged children.

Recyclable Waste. Refers to materials that can be recovered, processed, and reused in manufacturing, such as plastic bottles, glass, metals, and paper products. In this study, recyclable waste is one of the three output categories produced by the classification system.

Residual Waste. Refers to solid waste that remains after recyclable and biodegradable materials have been separated, consisting of materials that are no longer suitable for recovery or composting, including contaminated items that cannot be easily reclaimed. In this study, residual waste is one of the three output categories of the classification system.

Image Classification. The process of assigning a label or category to an image based on its visual content. In this study, image classification is used to assign a waste category – biodegradable, recyclable, or residual – to each waste image captured by the system.

MobileNetV2. A lightweight convolutional neural network architecture designed for efficient image classification tasks with reduced computational requirements. In this study, MobileNetV2 serves as the primary model used to classify waste images.

Multi-frame Prediction. Refers to the process of analyzing multiple image frames of the same object to generate classification results. In this study, multiple frames of a waste object are used to improve the reliability of the classification output.



Temporal Aggregation. The method of combining predictions obtained from multiple image frames over time to produce a more stable final result. In this study, temporal aggregation is used to reduce classification errors caused by noise or visual variability.

Waste Classification. The process of identifying and categorizing waste materials into predefined groups based on their characteristics. In this study, waste classification refers to the automated identification of solid waste as biodegradable, recyclable, or residual using image-based deep learning techniques.



CHAPTER TWO

REVIEW OF RELATED LITERATURE

2.1 Solid Waste Management and Waste Segregation Challenges

Effective solid waste management remains a significant challenge in many urban environments due to inadequate waste segregation practices and the contamination of recyclable materials. Studies have shown that improper sorting of waste at the source significantly reduces the efficiency of recycling systems and increases the operational cost of waste management facilities. Shreya et al. (2023) emphasized that the lack of public awareness and proper segregation practices is one of the major barriers to efficient recycling. Their review highlighted that contamination of recyclable waste often leads to the rejection of otherwise recyclable materials during processing.

Similarly, Arishi (2025) reported that many households fail to properly segregate waste due to limited knowledge and motivation, which results in mixed waste streams and reduced recycling efficiency. In their study, a deep learning-based waste detection system was proposed to improve classification accuracy and facilitate automated waste sorting. Pučnik et al. (2024) further demonstrated that contamination significantly affects waste sorting accuracy by comparing sensor-based and camera-based classification systems. Their findings indicated that contaminated plastic packaging often produces ambiguous visual signals, making automated classification more challenging.

Moreover, Dokl et al. (2025) conducted a systematic review of machine learning applications in waste management and concluded that automated classification systems have the potential to significantly improve recycling efficiency by reducing human error in waste sorting processes. However, they also noted that contamination and visual variability in waste materials remain major obstacles for automated classification systems. These findings highlight the need for more robust waste recognition approaches



capable of handling contaminated and visually complex waste objects. Consequently, image-based classification systems using deep learning models have been increasingly explored as potential solutions for improving waste segregation efficiency.

2.2 Waste Classification Standards and Categories

In the Philippines, the classification of municipal solid waste is governed by Republic Act No. 9003, also known as the Ecological Solid Waste Management Act of 2000, which mandates the segregation and collection of solid waste into three primary categories: biodegradable, recyclable, and residual (Republic Act No. 9003, 2001). This three-category framework provides a standardized basis for waste segregation and disposal at the source and is implemented across local government units, public institutions, and commercial establishments nationwide. Biodegradable waste refers to organic materials that can be decomposed by microorganisms, such as food scraps, yard waste, and other putrescible materials. Recyclable waste comprises materials that can be recovered and processed for reuse in manufacturing, including plastic bottles, glass, metals, and paper products. Residual waste encompasses materials that remain after the removal of biodegradable and recyclable fractions and are no longer suitable for recovery, including contaminated or soiled waste items that cannot be easily reclaimed.

The adoption of this standardized three-category classification framework is particularly important in the development of automated waste recognition systems. Several studies have emphasized that aligning automated classification frameworks with established waste management standards improves the practical relevance and deployability of technology-based solutions (Dokl et al., 2025; Shreya et al., 2023). Furthermore, many publicly available waste image datasets used in training deep learning models are organized around similar broad waste categories, making it feasible to develop classification systems using internet-sourced data while remaining applicable to standard waste segregation practices (Nahiduzzaman et al., 2025; Arishi, 2025).

The use of broader functional waste categories—biodegradable, recyclable, and residual – also reduces the complexity of the classification task compared to fine-grained material-level categorization, while still providing actionable information for waste



sorting. Gurcan and Soylu (2025) noted that the granularity of waste classification categories directly influences dataset requirements and model training complexity. By adopting the waste classification categories mandated under Philippine environmental legislation, the present study ensures that its classification framework is grounded in a legally recognized standard, supports compliance with national solid waste management policy, and enables the proposed system to be integrated into existing institutional waste management practices.

2.3 Image-Based Waste Classification Systems

Recent advancements in computer vision and deep learning have enabled the development of image-based waste classification systems capable of automatically identifying waste materials. These systems utilize digital images of waste objects and apply machine learning algorithms to classify them into predefined categories. Nahiduzzaman et al. (2025) proposed a hierarchical deep learning framework for waste classification using the TriCascade WasteImage dataset, which contains over 35,000 labeled images collected from multiple sources. Their results demonstrated that large-scale datasets combined with convolutional neural networks can achieve classification accuracy exceeding 95%.

Other researchers have explored object detection architectures to perform waste classification in real-time environments. Dipo et al. (2025) introduced a YOLO-based detection model trained on a custom dataset of waste images to perform simultaneous detection and classification of waste materials. Similarly, Rajeev et al. (2025) investigated the performance of different YOLO architectures for electronic waste classification and found that newer models such as YOLOv8 achieved higher accuracy and faster inference times compared to earlier versions.

In addition to detection-based approaches, several studies have explored hybrid classification frameworks that combine multiple models. Yang et al. (2023) proposed a system that integrates both image classification and object detection models, where predictions from different models are combined using a voting-based fusion algorithm. Their results showed that combining multiple model predictions improved classification



accuracy compared to individual models. Furthermore, Ling et al. (2026) introduced a modified YOLO architecture designed specifically for waste classification tasks, which achieved improved precision by incorporating additional feature fusion modules.

Although these studies demonstrate the potential of image-based classification systems for waste sorting, most of them rely on single-image predictions. In real-world environments, waste objects may appear under varying conditions such as different orientations, lighting conditions, or contamination levels. These factors may cause classification inconsistencies when only a single image is analyzed. Therefore, approaches that incorporate multiple observations of the same object may help improve classification reliability.

2.4 Deep Learning Approaches for Image Classification

Deep learning has become one of the most effective approaches for image classification tasks due to its ability to automatically learn hierarchical features from large datasets. Convolutional neural networks (CNNs) have demonstrated significant improvements in classification performance compared to traditional machine learning techniques. Basheer et al. (2023) compared several deep learning architectures for recyclable waste classification and found that pre-trained convolutional neural networks such as DenseNet, MobileNetV2, and ResNet significantly outperformed conventional machine learning models.

Transfer learning has also been widely adopted to improve classification accuracy when training data is limited. Gurcan and Soylu (2025) conducted a comparative analysis of multiple CNN architectures using transfer learning techniques and reported that deeper networks achieved higher classification accuracy, while lightweight models provided faster training and inference times. Similarly, Ghazal et al. (2025) developed a deep learning-based waste classification system using a large dataset containing more than 15,000 labeled waste images. Their results demonstrated that deep neural networks can achieve classification accuracy exceeding 98% when trained on sufficiently large datasets.



Another approach involves combining multiple deep learning models to improve classification performance. Pitakaso et al. (2024) proposed an ensemble-based classification system that integrates segmentation and multiple CNN models for waste recognition in disaster waste environments. The ensemble method improved classification accuracy compared to individual models, demonstrating the potential of combining multiple predictions to enhance recognition performance. Despite these advancements, most deep learning approaches rely on analyzing a single image of the waste object, which may not fully capture variations in object appearance. This limitation suggests the need for classification methods that consider multiple observations of the same object.

2.5 Convolutional Neural Networks for Visual Recognition

Convolutional neural networks serve as the fundamental architecture for most modern image recognition systems. CNNs utilize convolutional layers to extract hierarchical features from images, enabling them to recognize complex patterns such as shapes, textures, and object boundaries. These networks have been widely applied in object recognition, medical imaging, and industrial automation due to their ability to automatically learn relevant features from raw image data.

Zhao et al. (2022) demonstrated the effectiveness of CNNs in waste classification by developing a compact neural network architecture combined with preprocessing techniques to improve classification accuracy. Their system employed adaptive image brightening and background cropping to reduce the effects of shadows and uneven lighting conditions. The results showed that CNN-based systems can achieve high classification accuracy even with relatively small model architectures when appropriate preprocessing techniques are applied.

Despite their effectiveness, CNN-based systems may still face challenges when dealing with complex real-world conditions such as varying illumination, occlusion, and object deformation. These challenges highlight the importance of selecting appropriate network architectures and preprocessing methods to ensure reliable performance in practical waste classification applications.



2.6 Lightweight Convolutional Neural Network Models

While deep neural networks can achieve high classification accuracy, many architectures require significant computational resources, making them unsuitable for real-time or embedded applications. To address this limitation, lightweight convolutional neural networks have been developed to reduce computational complexity while maintaining acceptable accuracy.

Gurcan and Soylu (2025) evaluated several lightweight CNN architectures, including MobileNetV2 and SqueezeNet, and reported that these models achieved competitive classification performance while requiring significantly fewer computational resources. Similarly, Naznine et al. (2025) proposed a lightweight CNN framework using depthwise separable convolutions and an extreme learning machine classifier, achieving high classification accuracy with approximately one million model parameters.

These findings demonstrate that lightweight neural network architectures can provide an effective balance between classification accuracy and computational efficiency. As a result, such architectures are widely used in applications that require real-time image processing or deployment on resource-constrained devices.

2.7 MobileNetV2 Architecture for Image Classification

Among lightweight CNN architectures, MobileNetV2 has gained significant attention due to its efficient design and strong performance in image classification tasks. Sandler et al. (2018) introduced the MobileNetV2 architecture, which incorporates inverted residual blocks and linear bottleneck layers to enhance feature representation while maintaining low computational cost. Furthermore, MobileNetV2 demonstrates an effective trade-off between accuracy and efficiency, achieving 72.0% top-1 accuracy on ImageNet with only 3.4 million parameters and 300 million multiply-add operations, while a larger variant reaches 74.7% accuracy with 6.9 million parameters. These results highlight the model's ability to deliver competitive performance while significantly reducing computational and memory requirements, making it highly suitable for mobile and embedded vision applications.



Several studies have demonstrated the effectiveness of MobileNetV2 for waste classification applications. Yong et al. (2023) applied MobileNetV2 to classify domestic waste images and reported improved classification accuracy compared to traditional CNN models. Similarly, Nzeuga et al. (2025) fine-tuned MobileNetV2 using transfer learning techniques for waste classification tasks, achieving high classification accuracy while maintaining efficient computational performance.

These findings indicate that MobileNetV2 is well-suited for applications requiring efficient image classification. Its lightweight architecture allows it to process images quickly while maintaining competitive accuracy, making it an appropriate choice for real-time waste classification systems.

2.8 Temporal Aggregation Methods for Prediction Stability

Although deep learning models can achieve high classification accuracy, predictions based on a single image may still be affected by noise, occlusion, or variations in object appearance. Temporal aggregation methods have been proposed to improve prediction stability by combining multiple predictions obtained from sequential observations.

Yang et al. (2023) demonstrated that combining predictions from multiple models using a voting-based fusion method improved classification accuracy in waste recognition tasks. Their findings suggest that aggregating multiple predictions can reduce classification errors caused by ambiguous or noisy input images.

Similarly, research in video-based object recognition has shown that analyzing multiple frames of the same object can improve classification consistency by capturing different visual perspectives. By incorporating temporal information from sequential frames, classification systems can reduce prediction fluctuations and improve overall reliability. These findings support the use of multi-frame prediction aggregation techniques in image-based classification systems.



2.9 Visual Variability in Real-World Waste Images

Real-world waste images often exhibit significant variability due to differences in lighting conditions, object orientation, background clutter, and contamination. Such variations can significantly affect the performance of image classification systems. Zhao et al. (2022) demonstrated that lighting variations and shadow effects can negatively impact classification accuracy, emphasizing the importance of preprocessing techniques to normalize image conditions.

In addition to lighting variability, contamination can also alter the visual appearance of waste objects. Miski et al. (2023) investigated the detection of contaminated plastic bottles and found that the presence of liquid residue can obscure key visual features used for classification. These challenges highlight the need for classification systems that are robust to environmental variability and visual noise.

Considering these challenges, incorporating multiple observations of the same object may help improve classification reliability. By capturing several frames from different angles or lighting conditions, classification systems can obtain more comprehensive visual information, reducing the impact of visual variability on prediction accuracy.

2.10 Image Dataset Preparation for Waste Classification

The performance of deep learning models is highly dependent on the quality and diversity of the training dataset. Proper dataset preparation, including image collection, annotation, and augmentation, plays a critical role in improving model generalization and classification accuracy.

Nahiduzzaman et al. (2025) demonstrated that large-scale datasets significantly improve classification performance by introducing the TriCascade WasteImage dataset containing more than 35,000 labeled images. The study highlighted the importance of dataset diversity in capturing different waste types and environmental conditions.

Similarly, Arishi (2025) emphasized the role of data augmentation techniques in improving model performance. By applying image transformations such as rotation,



flipping, cropping, and brightness adjustments, the dataset size was effectively increased, allowing the model to learn more robust visual features. These techniques help prevent overfitting and improve classification performance when training data is limited.

Proper dataset preparation is therefore essential for developing reliable waste classification systems. Combining diverse datasets with augmentation techniques allows deep learning models to better handle variations in real-world waste images.

2.11 Early Childhood Environmental Education

The behavioral roots of solid waste mismanagement are, in large part, a consequence of inadequate environmental education during early childhood — a period that developmental and environmental education researchers have long identified as critically sensitive for the formation of pro-environmental attitudes and habits. Barraza (2006) was among the first to systematically document that children between the ages of 6 and 12 exhibit significantly greater openness to environmental value formation than older learners, as their cognitive and moral frameworks are still in active development and highly susceptible to experiential influence. This finding has been corroborated by Cheng and Monroe (2012), who demonstrated in a large-scale study of elementary students in the United States that children with higher nature connectedness and environmental self-efficacy — both of which are cultivable through school-based environmental programs — were substantially more likely to exhibit environmentally responsible behaviors, including waste reduction and recycling, in their daily lives.

In the Philippine context, where RA 9003 mandates source segregation yet compliance remains low, early childhood environmental education takes on particular policy relevance. Rickinson et al. (2004) conducted a comprehensive review of outdoor and school-based environmental learning programs and concluded that experiential, activity-based environmental education significantly outperforms traditional classroom instruction in producing durable knowledge retention and behavioral change in children. Their review further noted that the effectiveness of environmental education interventions is strongly mediated by the degree to which children can directly manipulate and interact



with the materials or concepts being taught, rather than merely observing or listening. This finding is central to the pedagogical rationale of the present study: by providing a physical device that children can interact with directly — presenting a waste object and immediately receiving a machine-generated classification — the prototype operationalizes the principles of activity-based environmental learning within the specific domain of RA 9003 waste segregation.

The role of children as agents of behavioral change within their households and communities further amplifies the significance of early childhood environmental education interventions. Kollmuss and Agyeman (2002) noted that the attitudes and behaviors cultivated in children frequently propagate outward into their families through a process of reverse socialization, wherein children influence the environmental behaviors of parents and siblings. Ernst and Monroe (2004) observed a similar dynamic in school-based environmental programs, finding that participating children not only modified their own behavior but also reported initiating conversations about waste and recycling practices within their households. These findings suggest that an educational prototype targeting elementary-aged children in Philippine schools and communities functions not merely as a classroom device but as a vector for broader community-level behavioral change, extending its impact well beyond the individual child learner.

2.12 Experiential Learning via Physical Interactive Prototypes

The use of physical, interactive prototypes as educational tools is driven by the principle that knowledge is most effectively constructed through active, hands-on engagement and immediate feedback. In the context of elementary environmental education, this learning process is engaged when a child physically presents a waste object to a classification device, observes the automated output, reflects on whether the result matches their expectation, and adjusts their mental model of waste categories accordingly—a process that traditional lecture-based instruction on waste segregation does not afford.



The specific value of physical, tangible prototypes over screen-based or purely verbal instructional media for young learners has been examined in the educational technology literature. Bers et al. (2014) demonstrated in a study of tangible programming environments for early childhood that children as young as five years old exhibited significantly deeper conceptual understanding and longer retention when learning was mediated by physical objects with immediate functional feedback, as compared to equivalent screen-based learning activities. The authors attributed this advantage to the multimodal engagement afforded by physical interaction: the simultaneous activation of visual, tactile, and motor pathways during hands-on learning produces stronger memory encoding than unimodal auditory or visual-only instruction.

Critically, the value of physical interactive prototypes in this context is independent of gamification. Resnick and Rosenbaum (2013) distinguished between tools that make learning fun through extrinsic rewards and game mechanics, and tools that make learning meaningful through genuine task engagement. The former approach, while motivating in the short term, does not reliably produce the conceptual depth required for durable behavioral change. The latter approach, exemplified by tools that give children real agency over real tasks, produces what the authors termed "learning through making." The waste classification prototype developed in this study is aligned with this second paradigm: it does not incorporate points, levels, or competitive elements, but instead presents waste classification as a genuine, consequential task in which the child's interaction with a physical waste object produces a real, machine-generated result. This task authenticity is expected to produce more durable learning outcomes than a gamified equivalent, as it more closely approximates the real-world behavior of waste segregation at the bin.

In the Philippine elementary school context, the deployment of physical interactive prototypes as learning materials is further motivated by the constraints of the local educational environment. Many public elementary schools operate with limited access to multimedia technology, and screen-based educational software requires ongoing digital infrastructure that is frequently unavailable in lower-resource settings. A



standalone hardware prototype that operates independently of internet connectivity or complex software environments presents a more equitable and robust instructional tool for the widest range of Philippine school contexts.

2.13 Human-Computer Interaction (HCI) Considerations for Child-Facing Automated Systems

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The specific value of physical, tangible prototypes over screen-based or purely verbal instructional media for young learners has been examined extensively in modern educational technology literature. Bers et al. (2014) demonstrated in a study of tangible programming environments for early childhood that children as young as five years old exhibited significantly deeper conceptual understanding and longer retention when learning was mediated by physical objects with immediate functional feedback, as compared to equivalent screen-based learning activities. The authors attributed this advantage to the multimodal engagement afforded by physical interaction: the simultaneous activation of visual, tactile, and motor pathways during hands-on learning produces stronger memory encoding than unimodal auditory or visual-only instruction. This finding is consistent with the broader literature on embodied cognition, which holds that physical interaction with the learning environment is not merely a pedagogical convenience but a constitutive component of concept formation in children (Wilson, 2002).



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2.14 Synthesis

The reviewed literature collectively establishes two bodies of knowledge that, while individually well-developed, have not been integrated into a single locally grounded system: the technical domain of deep learning-based waste image classification, and the pedagogical domain of experiential environmental education for children.



Within the technical domain, the convergence of evidence is clear. Lightweight convolutional neural network architectures, and MobileNetV2 in particular, have demonstrated consistent suitability for real-time waste image classification by balancing classification accuracy with the computational efficiency required for edge deployment (Sandler et al., 2018; Nzeuga et al., 2025; Gurcan & Soylu, 2025). Transfer learning from large-scale image corpora such as ImageNet substantially reduces the data requirements for effective waste classification, enabling competitive performance even on modestly sized datasets (Basheer et al., 2023; Ghazal et al., 2025). The persistent challenge across this body of literature is prediction instability under real-world disposal conditions, where individual image frames may be degraded by contamination, occlusion, or variable lighting. Temporal aggregation techniques — specifically majority voting across multiple independent observations of the same object — have been proposed and validated as a principled mechanism for reducing this frame-level noise and producing more stable final predictions (Yang et al., 2023; Nguyen et al., 2024). The five-frame majority voting scheme adopted in the present study is directly grounded in this literature. The classification categories used — biodegradable, recyclable, and residual — are anchored in the legally recognized framework of Republic Act No. 9003 (2001), ensuring that the technical system's outputs are directly actionable within Philippine waste management practice (Dokl et al., 2025; Shreya et al., 2023).

Within the pedagogical domain, the evidence is equally convergent. Early childhood — specifically the elementary school years spanning ages 6 to 12 — constitutes the most developmentally sensitive period for the formation of pro-environmental attitudes and behavioral habits (Barraza, 2006; Cheng & Monroe, 2012). Traditional lecture-based and passive instructional modalities, which constitute the predominant approach to waste segregation education in Philippine elementary schools, have been repeatedly demonstrated to be insufficient for producing durable behavioral change in young learners (Rickinson et al., 2004; Kollmuss & Agyeman, 2002). Experiential, activity-based environmental education — particularly that which involves direct physical manipulation of materials and immediate, automated feedback — significantly outperforms passive instruction in both knowledge retention and long-term



behavioral transfer (Bers et al., 2014; Resnick & Rosenbaum, 2013). The literature further establishes that children who acquire correct waste segregation behaviors through experiential means function as agents of reverse socialization within their households, extending the behavioral impact of educational interventions beyond the individual child (Ernst & Monroe, 2004). Human-computer interaction research on child-facing systems additionally specifies that interactive devices for young users must prioritize simplicity, immediacy of feedback, and physical tangibility over screen-based or gamified interfaces (Druin, 2002; Read & Bekker, 2011; Fails & Druin, 2003).

Despite the maturity of both bodies of literature, a critical and underexplored gap persists at their intersection. While AI-based waste classification systems have been developed for industrial and facility-level deployment, and while experiential environmental education has been validated as effective for young learners, no locally developed, hardware-based interactive prototype currently exists that integrates MobileNetV2-class multi-frame classification with a child-accessible physical interface designed specifically to teach RA 9003 waste segregation categories to Philippine elementary-aged children. Existing systems address either the technical sorting problem or the educational engagement problem in isolation. The behavioral root cause of waste mismanagement in the Philippines — the absence of deeply ingrained segregation habits formed in early childhood — remains unaddressed by technology-centric solutions designed for adult or industrial users.

This study directly fills that gap by designing, implementing, and demonstrating a physical interactive educational prototype that deploys a MobileNetV2-based five-frame majority voting classification pipeline on an edge device, producing immediate, unambiguous hardware feedback mapped to the three RA 9003 categories for elementary-aged children. The prototype operationalizes experiential learning theory within the specific domain of RA 9003 waste segregation, functioning as a non-gamified, task-authentic instructional device that engages the sensorimotor and cognitive learning pathways most active during early childhood and that has the practical potential to



propagate correct segregation behavior from individual child learners into their households and communities.

2.15 Theoretical Framework

The present study is grounded in two complementary theoretical pillars that jointly account for its dual nature as both an engineering system and an educational intervention. The first pillar provides the theoretical foundation for the AI classification pipeline; the second provides the theoretical foundation for the hardware interface and its intended learning effects on child users.

Deep Learning and Hierarchical Feature Extraction Theory

The technical operation of the classification system is grounded in the theory of deep learning and hierarchical feature extraction, which explains how multi-layer convolutional neural networks acquire the capacity to categorize visual inputs from raw pixel data. Convolutional neural networks learn hierarchical feature representations by applying successive layers of learned filters that detect progressively abstract visual structures: early layers encode low-level attributes such as edges, color gradients, and textures, while deeper layers encode higher-order semantic patterns such as object contours, material surfaces, and category-distinguishing shapes (LeCun et al., 1998). This hierarchical representation enables a trained network to generalize from labeled training examples to previously unseen instances of the same visual category.

Transfer learning extends this theoretical basis to settings where labeled domain-specific training data is limited. By initializing a model with weights pre-trained on a large-scale general image corpus — specifically ImageNet in the present study — the convolutional base arrives at the waste classification task already encoding rich, reusable visual representations that can be adapted through fine-tuning to the specific visual signatures of biodegradable, recyclable, and residual waste categories (Sandler et al., 2018; Gurcan & Soylu, 2025). The MobileNetV2 architecture operationalizes this theory through its inverted residual blocks and linear bottleneck layers, which preserve the representational



capacity required for effective feature extraction while minimizing the multiply-add operations required for each forward pass — a property that is essential for real-time, multi-frame inference on an edge device (Sandler et al., 2018).

The five-frame temporal aggregation mechanism derives its theoretical justification from the probabilistic interpretation of ensemble prediction: when multiple independent observations of the same object are classified and their predictions aggregated by majority vote, the influence of frame-specific noise — including contamination artifacts, partial occlusion, and lighting variation — is statistically attenuated, and the resulting consensus prediction more reliably reflects the true visual category of the object (Yang et al., 2023; Nguyen et al., 2024). Together, these theoretical principles establish that the classification component of the prototype is capable of producing accurate, stable, and real-time waste category labels from a continuous camera feed.

Experiential and Tangible Learning Theory

The educational interface of the prototype is grounded in experiential learning theory, as formulated by Kolb (1984) and traceable to the pragmatist philosophy of Dewey (1938), which holds that durable knowledge is not transmitted through passive reception but is constructed through a cyclical process of direct concrete experience, reflective observation, abstract conceptualization, and active experimentation. In the context of the present prototype, this learning cycle is engaged when a child presents a physical waste object to the device (concrete experience), observes the machine-generated classification output (reflective observation), forms or revises a mental model of which waste categories specific objects belong to (abstract conceptualization), and subsequently tests that model by presenting additional objects (active experimentation). Critically, this complete cycle is unavailable through the lecture-based and printed-visual-aid instruction that currently dominates waste segregation education in Philippine elementary schools, as passive modalities do not provide the immediate, object-specific feedback that the experiential cycle requires.



Tangible learning theory — operationalized through the educational technology literature on physical interactive prototypes for young children — extends this foundation by specifying why physical, hardware-mediated interaction produces superior learning outcomes compared to equivalent screen-based instruction for the 6-to-12 age range. Bers et al. (2014) demonstrated that children learning through physical objects with immediate functional feedback simultaneously activate visual, tactile, and motor processing pathways, producing multimodal memory encoding that is substantially stronger than the unimodal encoding produced by visual-only or auditory instruction. Wilson's (2002) embodied cognition framework further establishes that physical interaction with the learning environment is not merely a pedagogical enhancement but a constitutive component of concept formation in developing children, as motor engagement with objects shapes the cognitive schemata through which those objects are subsequently categorized.

Resnick and Rosenbaum (2013) contribute the critical distinction between gamified educational tools and task-authentic educational tools: the former employ extrinsic reward mechanisms to sustain engagement but produce shallow conceptual encoding, while the latter — which present real tasks with real consequences — produce what the authors describe as "learning through making," characterized by deeper understanding and greater behavioral durability. The prototype developed in this study is explicitly designed as a task-authentic tool: it presents waste classification as a genuine real-world task, its output is a real machine-generated classification, and no points, levels, or competitive elements are involved. This alignment with the task-authentic paradigm is expected to produce more durable learning outcomes and, through the reverse socialization mechanism described by Kollmuss and Agyeman (2002), to propagate correct waste segregation behavior from the child into the household environment.

Together, these two theoretical pillars establish the interdisciplinary scientific basis of the study. Deep learning and feature extraction theory accounts for why and how the system can accurately and stably classify contaminated waste in real time; experiential and tangible learning theory accounts for why and how the hardware interface translates that



classification capability into a meaningful and durable learning experience for elementary-aged children.

2.16 Conceptual Framework

The conceptual framework situates the foregoing technical methodology within the broader educational and environmental objectives of the study. It is organized according to an Input–Process–Output (IPO) model, as illustrated in Figure 3.4, and provides a systematic representation of the transformation sequence through which raw physical and digital inputs are converted, via the AI classification pipeline and hardware signal processing, into an experiential learning event for the child user.

Input. Three concurrent inputs initiate each classification cycle: the student's categorical guess registered via one of the four push buttons (biodegradable, recyclable, residual, or others); the waste image, captured as five RGB frames via the camera module; and the system activation signal triggered by the button press.

Process. The process stage encompasses four sequential operations. First, the image acquisition module captures five consecutive frames and preprocesses each to 224×224 pixels with pixel normalization. Second, the MobileNetV2 model performs feature extraction on each frame and generates a probability distribution over the four waste categories. Third, the multi-frame majority voting mechanism—with confidence-based tie-breaking—aggregates the five predictions into a single authoritative classification result. Fourth, the system compares the AI-determined classification against the student's button-press guess to determine correctness.

Output. The process stage produces three simultaneous outputs: a correct classification result across the four categories; physical segregation of the waste item into the appropriate bin via the rotating platform and drop gate; and real-time multimodal reinforcement feedback delivered through the LCD screen, LED strip, and buzzer. The cumulative effect of repeated interactions across these output channels is the primary outcome of interest: increased waste segregation literacy in the child user, operationalized



through pre-test versus post-test knowledge gains, improved waste category identification accuracy, and the internalization of correct segregation habits consistent with RA 9003.

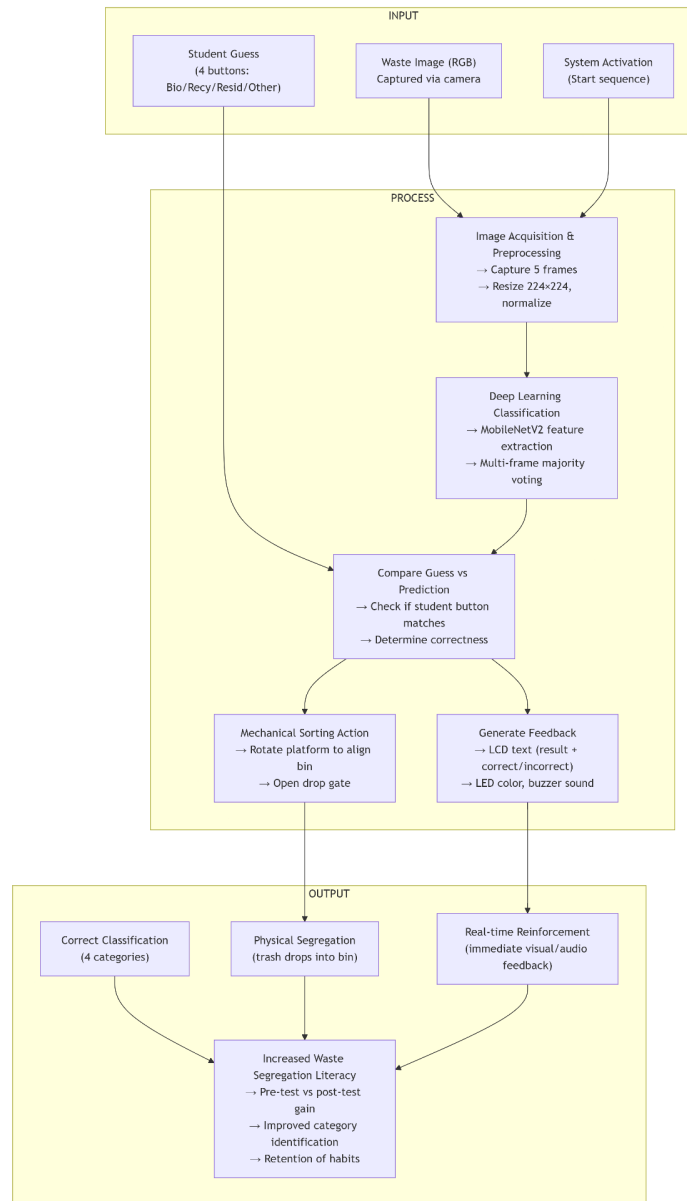


Figure 2.1: Conceptual Framework (Input-Process-Output)]



CHAPTER THREE

METHODOLOGY

3.1 Research Design

This study employs an experimental and quantitative research design focused on the development, implementation, and evaluation of a deep learning-based waste classification system integrated into an interactive educational hardware prototype. The experimental nature of the research arises from the construction of a functional classification system through specific computational methods, controlled training procedures, and systematic testing under defined conditions. The quantitative dimension is reflected in the use of numerical performance metrics—including classification accuracy, precision, recall, and F1-score—to objectively evaluate and compare system performance.

The study involves training the MobileNetV2 convolutional neural network on publicly available waste image datasets representing four waste categories: biodegradable, recyclable, residual, and others. These categories align with the solid waste classification framework mandated by Republic Act No. 9003 (2001) and are directly applicable to practical waste segregation contexts. The inclusion of an "Others" category is a deliberate design decision intended to accommodate unrecognizable or ambiguous items and to prevent the system from forcing a false positive assignment among the three primary categories. The use of publicly available datasets ensures reproducibility and eliminates constraints associated with custom image collection, which was not feasible within the study's timeframe.

A central element of the research design is the comparative evaluation between single-frame and multi-frame classification. The single-frame approach, in which only



one image of a waste object is used for classification, serves as the baseline condition. The multi-frame approach, in which five consecutive images of the same object are independently classified and their predictions aggregated through a majority voting mechanism with a confidence-based tie-breaker, represents the proposed innovation of this study. By systematically comparing these two conditions under identical dataset and model configurations, the research design directly addresses the core problem of prediction instability caused by contamination and visual variability in waste images (Yang et al., 2023; Nguyen et al., 2024). The comparative results provide measurable evidence of the extent to which multi-frame aggregation improves classification reliability.

3.2 Research Procedures

The methodology of this study follows a structured and sequential workflow that progresses logically from raw input data to a fully evaluated and physically deployed classification system. The procedure is organized into five phases: dataset preparation, model configuration and training, multi-frame prediction aggregation, performance evaluation, and comparative analysis of single-frame versus multi-frame results. The entire software pipeline is implemented in Python using established deep learning and computer vision frameworks. Each phase builds upon the outputs of the preceding phase, ensuring a coherent and reproducible research pipeline that is directly connected to the study's research questions and objectives.

3.2.1 System Architecture

The overall architecture of the proposed system is composed of input components, a central processing unit, and output components, as illustrated in Figure 3.1.

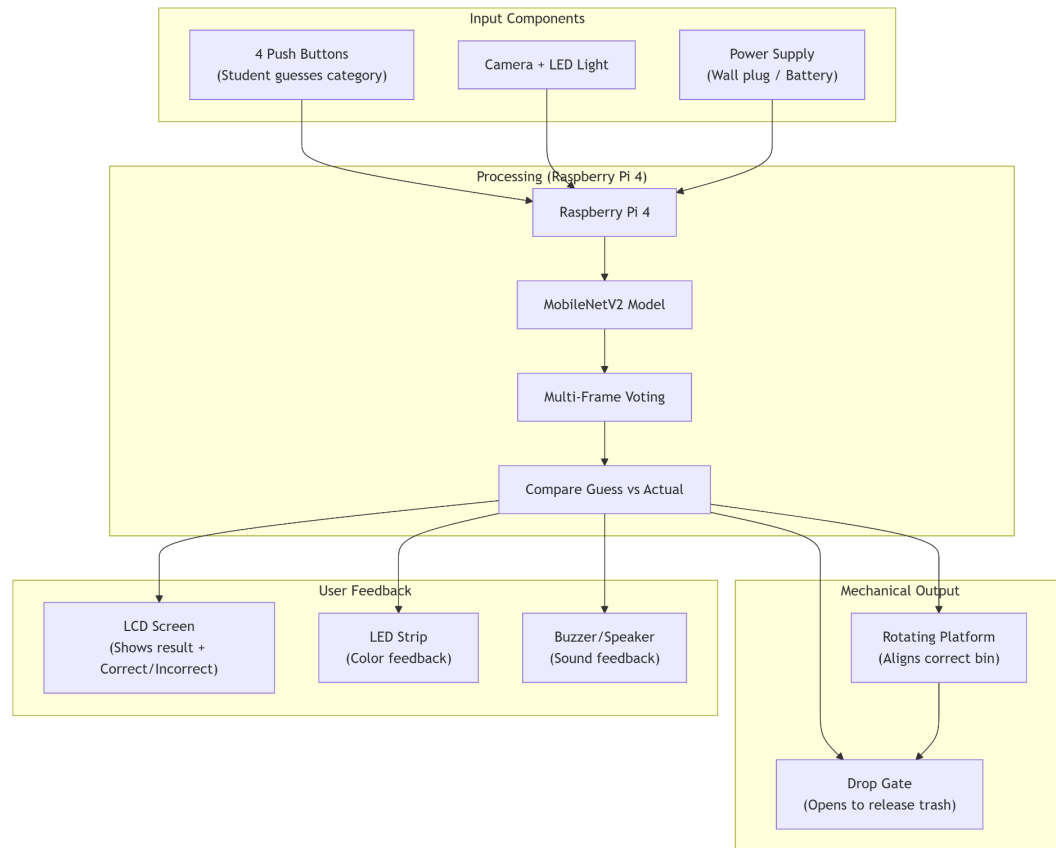


Figure 3.1: System Architecture Diagram

Input to the system is provided through three channels. Four push buttons allow the student user to register a categorical guess—selecting from among the biodegradable, recyclable, residual, or others categories—prior to the system's automated assessment. A camera module, mounted in the top area of the enclosure and illuminated by an integrated LED light for consistent exposure, captures image frames of the waste object placed on the platform. Power is supplied via either a wall plug or battery pack, providing operational flexibility for both classroom and field deployment contexts.

All input signals are routed to a **Raspberry Pi 4**, which serves as the central processing unit and executes the entire classification pipeline in real time. The Raspberry Pi 4 runs the MobileNetV2 deep learning model and the multi-frame voting logic, and coordinates



all downstream output peripherals. The Raspberry Pi 4 is mounted at the rear of the physical enclosure, where it controls all connected components.

Upon completing the classification and comparison operations, the system generates two categories of output. The **User Feedback** output set comprises an LCD screen, an LED strip, and a buzzer/speaker. The LCD screen displays the classification result alongside a correct or incorrect verdict relative to the student's guess. The LED strip provides color-coded visual feedback—green for a correct guess, red for an incorrect one—and the buzzer/speaker emits a corresponding audio signal. The **Mechanical Output** set comprises a rotating platform, which aligns the correct waste bin beneath the drop gate, and the drop gate itself, which opens to release the waste item into the appropriate bin once the feedback sequence is complete.

3.2.2 Hardware Prototype Layout

The physical enclosure of the prototype is structured across four vertically arranged zones, as depicted in Figures 3.5

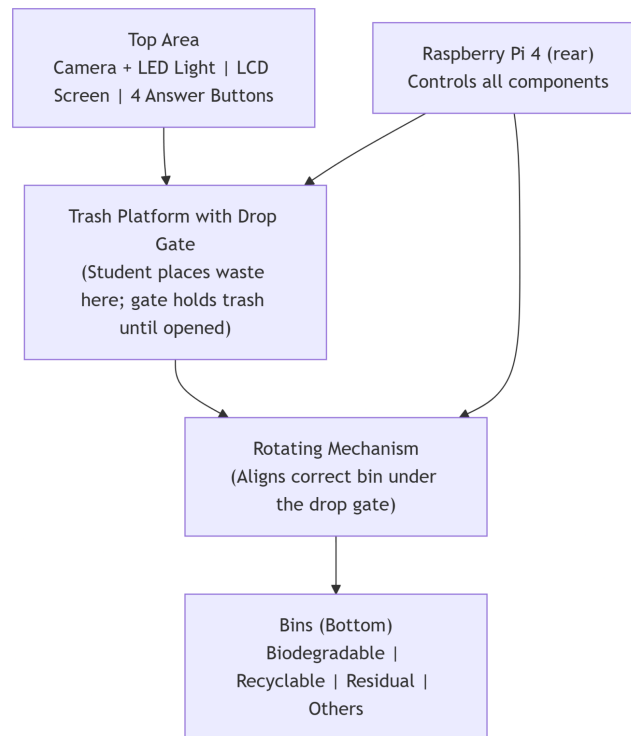


Figure 3.5.1: Hardware Prototype Layout

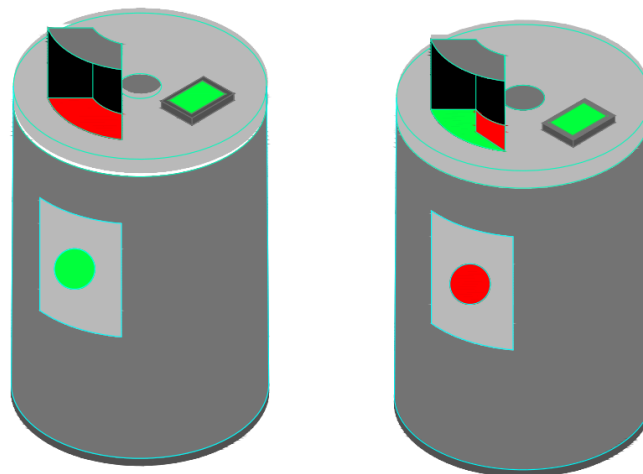


Figure 3.5.2: Hardware Prototype Design (Platform Closed & Opened)

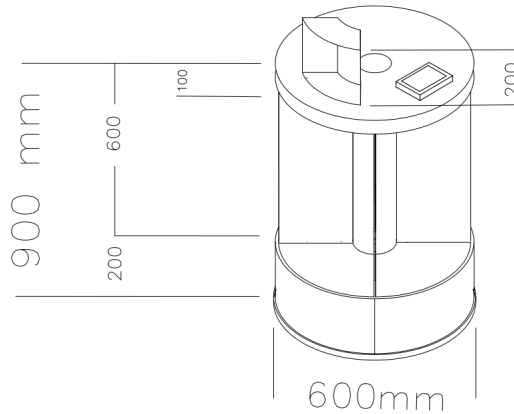


Figure 3.7: Hardware Prototype Design Dimensions

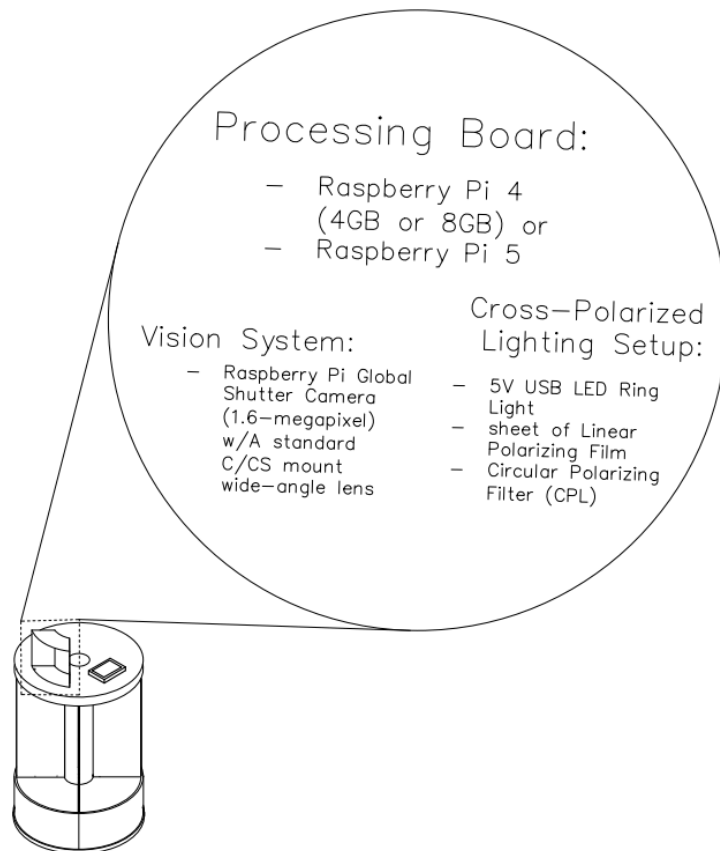


Figure 3.8: Hardware Prototype Details



The **top area** houses the primary user interface and sensing components: the camera module and LED illumination light for image capture, the LCD screen for displaying classification results and correctness feedback, and the four answer buttons through which the student submits a categorical guess.

The **middle zone** constitutes the **trash platform with drop gate**. The student places the waste object on this platform before initiating the assessment sequence. The gate mechanism retains the waste item on the platform until the classification and feedback cycle is fully completed, at which point it opens to release the item downward.

Below the platform, the **lower-middle zone** contains the **rotating mechanism**. This motorized assembly rotates to align the correct waste bin—corresponding to the AI-determined classification—directly beneath the drop gate prior to its opening, ensuring accurate physical waste routing.

The **bottom zone** houses four segregated **collection bins**, one for each waste category: biodegradable, recyclable, residual, and others. Waste items are deposited into the appropriate bin by gravity upon the opening of the drop gate.

The Raspberry Pi 4, mounted at the rear of the enclosure, governs all electromechanical components across these zones via its GPIO interface.

3.2.3 System Operation Flow

The experiential learning interaction cycle proceeds according to the sequence illustrated in Figure 3.2.

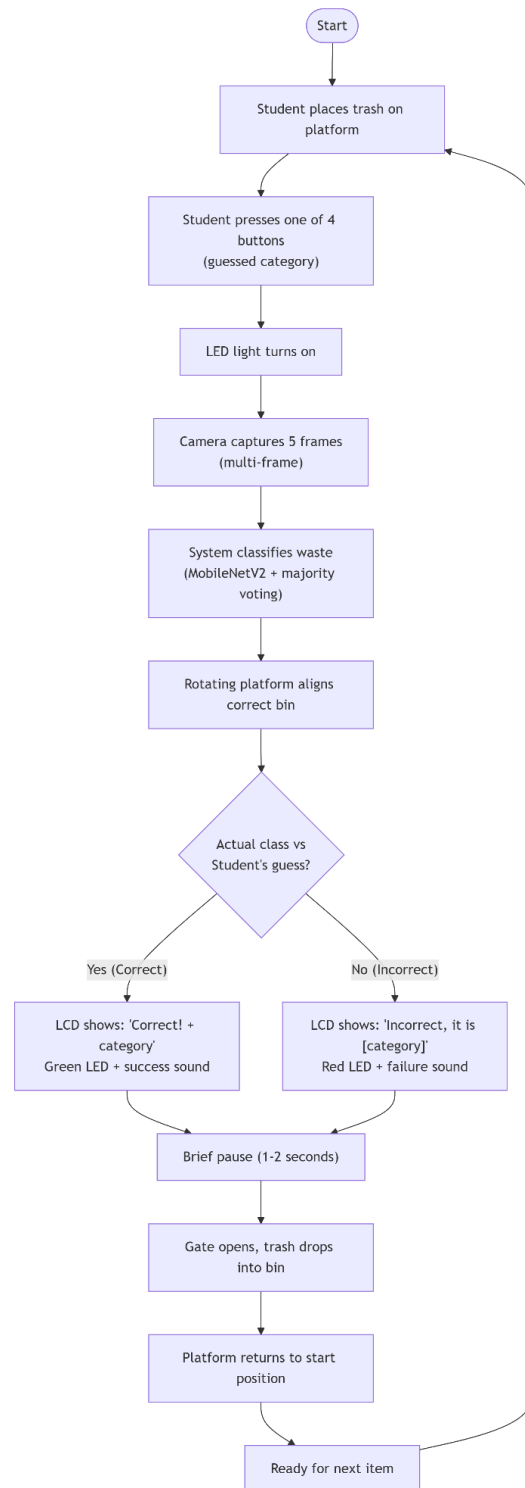


Figure 3.2: System Operation Flowchart



A single classification cycle begins when the student places a waste item on the platform. The student then presses one of the four buttons to register a categorical guess. This button press simultaneously activates the LED illumination light and triggers the image acquisition sequence, during which the camera captures five consecutive frames of the waste object.

Each captured frame is immediately resized and normalized, then independently classified by the MobileNetV2 model. The five resulting predictions are aggregated through the majority voting mechanism to produce the system's final classification output. Concurrently, the rotating platform begins to align the bin corresponding to the AI-determined class beneath the drop gate.

The system then compares the AI-determined classification against the student's button-press guess. If the student's guess matches the AI classification, the LCD screen displays a confirmation message identifying the correct category, the LED strip illuminates green, and the buzzer emits a success sound. If the guess does not match, the LCD displays a corrective message indicating the actual category, the LED strip illuminates red, and the buzzer emits a failure tone. This immediate multimodal feedback is the primary pedagogical mechanism of the prototype, providing the student with unambiguous, sensory-rich reinforcement aligned with the principles of experiential and tangible learning.

Following a brief pause of one to two seconds—sufficient for the student to read the feedback—the drop gate opens and the waste item falls into the correctly aligned bin. The platform then returns to its default starting position, and the system enters a ready state for the next item, completing the cycle.

3.2.4 Materials

The implementation of the proposed classification system requires a combination of hardware resources, software libraries, and image datasets. The primary processing hardware is a **Raspberry Pi 4** single-board computer, which serves as the deployment



target for the trained MobileNetV2 model and executes the entire real-time classification pipeline, multi-frame aggregation logic, and peripheral control operations. The Raspberry Pi 4 interfaces with a camera module for image capture, an LED light for illumination, an LCD screen, an LED strip, a buzzer/speaker, four push buttons, a stepper or servo motor driving the rotating platform, and a gate release actuator for the drop mechanism.

Model development, training, and evaluation are conducted on a personal computer (PC) equipped with a multi-core processor and, where available, a graphics processing unit (GPU) to accelerate the training of the deep learning model. GPU computation is particularly beneficial during training, as it significantly reduces the time required for gradient computation across large numbers of training images.

The software environment is built on Python, which serves as the primary programming language for all data processing, model implementation, and evaluation tasks. TensorFlow and Keras are used for constructing, training, and deploying the MobileNetV2 model, providing access to pre-trained ImageNet weights and a high-level API for defining custom classification layers. OpenCV handles image reading, resizing, and preprocessing operations. NumPy supports numerical array operations throughout the preprocessing and aggregation pipeline. Pandas is used for organizing evaluation results and managing dataset metadata. Matplotlib facilitates the visualization of training history, confusion matrices, and performance comparison charts. Scikit-learn provides implementations of the classification metrics used in the performance evaluation phase, including accuracy, precision, recall, F1-score, and confusion matrix computation.

The primary dataset used in this study is the TriCascade WasteImage dataset introduced by Nahiduzzaman et al. (2025), which contains over 35,000 labeled waste images collected from multiple publicly available sources. This dataset is selected for its large scale, label diversity, and the availability of waste images under varying environmental conditions, including contaminated and partially obscured samples. To supplement the primary dataset with additional examples of contaminated and real-world waste scenarios, the TACO (Trash Annotations in Context) dataset is incorporated as a



secondary source. The TACO dataset provides annotated images of waste objects captured in outdoor and natural environments, many of which exhibit realistic contamination, partial occlusion, and background clutter (Arishi, 2025). The combination of these two sources enables the model to be trained on a diverse and representative collection of waste images aligned with the four-category classification framework.

3.2.5 Dataset Preparation

Images from the TriCascade WasteImage and TACO datasets are organized into four class directories corresponding to the biodegradable, recyclable, residual, and others waste categories. Each image is assigned to its appropriate waste category based on the classification definitions established under Republic Act No. 9003 (2001). Biodegradable class images include organic waste items such as food scraps and plant matter. Recyclable class images include plastic bottles, metal containers, glass, and paper products. Residual class images include contaminated or composite waste items that cannot be classified as either recyclable or biodegradable. The others category encompasses items that do not correspond to any of the three primary waste classes—such as ambiguous or unrecognizable objects—thereby preventing the system from producing a forced false assignment. The resulting organized dataset provides a structured and labeled foundation for model training and evaluation.

The dataset is partitioned into three subsets using a 70/15/15 split ratio: 70% of images are allocated for training, 15% for validation, and 15% for testing. The training set is used to update the model's weights during the supervised learning process. The validation set is used to monitor training performance and guide decisions such as early stopping to prevent overfitting. The test set is held out entirely from the training and validation phases and is used only during the final performance evaluation to provide an unbiased estimate of classification accuracy and stability.

All images are resized to 224×224 pixels to match the input dimensions required by the MobileNetV2 architecture, and pixel values are normalized to the range $[0, 1]$ by dividing by 255. To address the variability introduced by contamination, differing lighting



conditions, and diverse object orientations—factors characteristic of real-world waste images—data augmentation is applied exclusively to the training set. Augmentation transformations include random horizontal and vertical flipping, rotation within a range of ± 30 degrees, brightness and contrast adjustments, zoom scaling, and shear transformations. Applied stochastically during training, these operations effectively expand the apparent diversity of the training data and improve the model's ability to generalize to unseen visual conditions (Arishi, 2025; Nahiduzzaman et al., 2025). The validation and test sets use only resizing and normalization to ensure that evaluation results reflect performance on unmodified images.

3.2.6 Model Configuration and Training

The classification model is built upon the MobileNetV2 architecture using a transfer learning strategy. MobileNetV2 is selected because its lightweight design—based on depthwise separable convolutions and inverted residual blocks—enables efficient feature extraction without the computational overhead associated with larger convolutional neural networks (Sandler et al., 2018). This efficiency is particularly important in the context of multi-frame classification, where the model must process five frames per object during inference on the resource-constrained Raspberry Pi 4. A computationally lightweight architecture ensures that multi-frame processing remains practical without introducing prohibitive latency.

The MobileNetV2 base model is initialized with weights pre-trained on the ImageNet dataset. Transfer learning is employed because the pre-trained base model has already learned rich hierarchical visual representations from a large and diverse image corpus, which can be effectively adapted to the task of waste classification even with a relatively limited training dataset (Gurcan & Soylu, 2025; Nzeuga et al., 2025). A custom classification head is appended to the base model, consisting of a global average pooling layer followed by a fully connected dense layer with softmax activation and **four output nodes** corresponding to the biodegradable, recyclable, residual, and others categories.



Training is conducted in two phases. In the first phase, the convolutional layers of the MobileNetV2 base model are frozen, and only the custom classification head is trained. This phase allows the classification layers to learn task-specific representations while preserving the general visual features encoded in the pre-trained base. In the second phase, the upper layers of the base model are unfrozen and the entire network is fine-tuned at a reduced learning rate. This fine-tuning phase allows the model to refine its feature representations in a manner more closely aligned with the visual characteristics of waste images, including the effects of contamination and surface variability.

The categorical cross-entropy loss function is used throughout training, as it is appropriate for multi-class classification tasks where each image is assigned to exactly one of the four waste categories. The Adam optimizer is applied due to its adaptive learning rate mechanism, which enables efficient convergence across the two-phase training process. An early stopping callback is implemented during both training phases to monitor validation loss and terminate training if no improvement is observed over a defined number of epochs, thereby preventing overfitting and reducing unnecessary computation.

3.2.7 Multi-Frame Prediction Aggregation

The multi-frame prediction aggregation mechanism constitutes the principal methodological contribution of this study and directly addresses the limitation of single-frame classification systems in handling contaminated and visually variable waste images. The fundamental premise of this approach is that a single image of a contaminated or partially occluded waste object may not provide sufficient or reliable visual information for accurate classification. When a classification model processes only one frame, transient visual artifacts—such as surface contamination, reflections, or partial occlusion—may dominate the image and produce an erroneous prediction (Miski et al., 2023; Zhao et al., 2022). By collecting and independently classifying multiple frames of the same object, the system obtains a broader representation of the object's visual



appearance, reducing the likelihood that any single adverse visual condition will determine the final output.

For each waste object presented to the system, five consecutive image frames are captured by the camera module. Each frame undergoes the same preprocessing pipeline—resizing to 224×224 pixels and pixel value normalization to the range [0, 1]—and is independently passed through the trained MobileNetV2 model. The model produces a class prediction and a corresponding probability distribution over the four waste categories for each frame. Once all five predictions are obtained, majority voting is applied: the waste category that receives the highest number of votes across the five frames is assigned as the final classification output for that object.

In cases where no single class achieves an absolute majority of three or more votes—that is, when two or more classes are tied in vote count—the tie-breaker procedure is activated. The system computes the average predicted confidence score for each tied class across all five frames and selects the class with the highest average confidence as the final output. This confidence-based tie-breaker ensures that the aggregation logic remains deterministic under all possible voting outcomes, while leveraging the model's probabilistic output to make the most reliable decision when vote counts are inconclusive.

The detailed frame-by-frame logic of this procedure is illustrated in Figure 3.3.

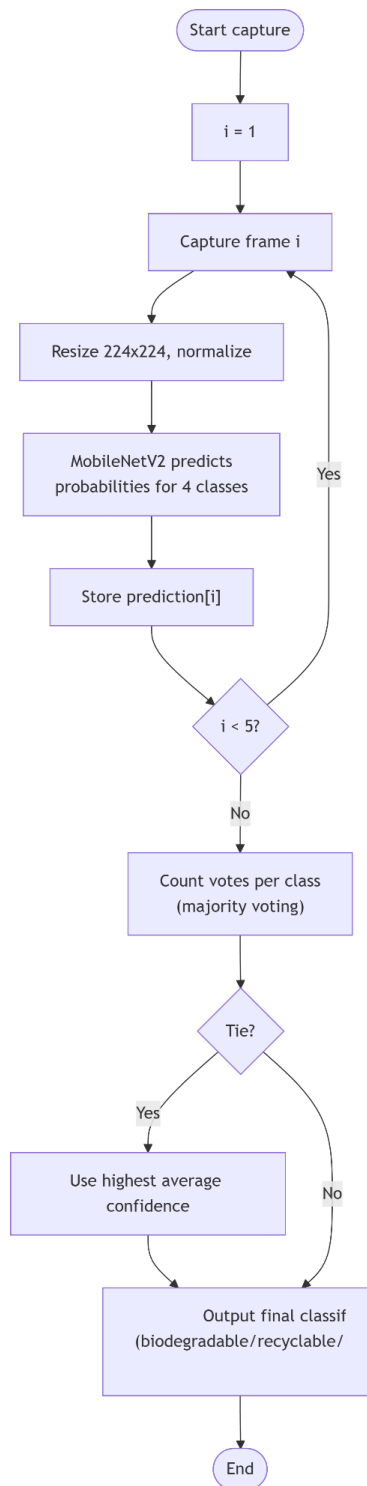


Figure 3.3: Multi-Frame Prediction Flowchart



The logic of the multi-frame aggregation procedure is also described in the following algorithm. For each waste object, five frames are acquired. Each frame is resized to 224×224 pixels and pixel values are normalized before being passed independently through the MobileNetV2 model to obtain a class prediction and its associated probability distribution. After all five frames have been processed, majority voting is applied across the five predictions. The class label receiving the majority of votes is returned as the final output. If no single class achieves an absolute majority of three or more votes, the class with the highest average predicted probability across all five frames is used as the tie-breaker.

Algorithm: Multi-Frame Prediction Aggregation

INPUT: waste_object, trained_model

OUTPUT: final_class_label

frames $\leftarrow$ acquire_frames(waste_object, N = 5)

predictions $\leftarrow$ []

confidences $\leftarrow$ { }

FOR each frame IN frames:

img $\leftarrow$ resize(frame, 224×224)

img $\leftarrow$ normalize(img)

probs $\leftarrow$ trained_model.predict(img) // vector of 4 class probabilities

pred $\leftarrow$ argmax(probs)

predictions.append(pred)

confidences[pred].append(max(probs))



END FOR

$\text{votes} \leftarrow \text{count_occurrences}(\text{predictions})$

$\text{final_class_label} \leftarrow \text{majority_vote}(\text{votes})$

IF $\text{tie_detected}(\text{votes})$:

$\text{final_class_label} \leftarrow \text{argmax}(\text{mean_confidence}(\text{confidences}, \text{tied_classes}))$

END IF

RETURN final_class_label

The choice of five frames as the aggregation window balances prediction stability with processing efficiency. With five independent predictions, majority voting requires at least three votes for a given class to determine the output, providing a meaningful degree of noise rejection without requiring a large number of frames that would significantly increase processing time on the Raspberry Pi 4. This frame count is consistent with temporal aggregation approaches used in sequential object recognition tasks, where a small number of observations has been found sufficient to substantially improve classification consistency (Nguyen et al., 2024). The majority voting mechanism, augmented by the confidence-based tie-breaker, ensures that the aggregation logic remains transparent and interpretable while achieving improved prediction stability over single-frame classification.

3.3 Performance Evaluation



The performance evaluation phase assesses the effectiveness of the proposed classification system using both single-frame and multi-frame prediction strategies. Evaluation is conducted on the held-out test set, which comprises 15% of the total dataset and was not used during any stage of model training or validation. For the single-frame condition, each test image is independently classified by the trained MobileNetV2 model and its prediction is compared directly against the ground-truth label. For the multi-frame condition, five frames drawn from the test set for each waste object are processed through the aggregation algorithm, and the majority-voted output is compared against the ground-truth label. This comparative structure allows the evaluation to measure not only the absolute performance of the system but also the specific improvement attributable to multi-frame aggregation.

Classification accuracy is the primary performance metric, defined as the proportion of correctly classified waste objects to the total number of objects in the test set. While accuracy provides a general indication of model performance, it is insufficient as a sole metric when class distributions may be uneven, or when different types of misclassification carry different practical implications. Classifying recyclable waste as residual, for instance, directly undermines material recovery efforts; classifying biodegradable waste as recyclable may contaminate recycling streams. For this reason, precision, recall, and F1-score are also computed for each class.

Precision measures the proportion of samples predicted as a given class that are actually members of that class, reflecting the system's tendency to produce false positives. Recall measures the proportion of actual members of a given class that are correctly identified, reflecting the system's ability to capture all instances of a category. The F1-score is the harmonic mean of precision and recall, providing a single balanced measure of per-class performance that is sensitive to both false positives and false negatives.

All four metrics—accuracy, precision, recall, and F1-score—are reported on a **per-class basis** for each of the four waste categories (biodegradable, recyclable, residual, and others), enabling independent assessment of the model's performance on each category. A



macro-averaged F1-score is additionally computed across all four classes to provide an overall indication of classification balance, giving equal weight to each category regardless of its frequency in the test set.

A confusion matrix is generated for both the single-frame and multi-frame evaluation conditions. The confusion matrix provides a complete breakdown of the model's classification decisions, showing for each true class how many instances were correctly classified and how many were misclassified into each of the other categories. Comparing the confusion matrices under single-frame and multi-frame conditions identifies specific class pairs where multi-frame aggregation most significantly reduces misclassification—particularly between visually similar classes such as contaminated recyclable materials and residual waste, or between ambiguous items and the others category.

In addition to these classification metrics, prediction stability is assessed by measuring the variance in class predictions across the five frames within each multi-frame group. A stable prediction is one in which all or nearly all five frames produce the same class label, while an unstable prediction is one in which frames produce conflicting labels. The proportion of waste objects for which all five frames agree on the correct class—referred to as full prediction consistency—is reported as a supplementary indicator of the reliability improvement attributable to the multi-frame approach. This metric directly quantifies the core advantage of the proposed system over single-frame baselines: by aggregating multiple predictions, the system produces outputs that are not only more accurate on average but also more consistent and reliable across a range of visual conditions introduced by contamination and environmental variability.



CHAPTER FOUR

RESULTS AND DISCUSSION

4.1 Overview

This chapter presents the results of the experiments conducted to develop and evaluate the proposed multi-frame image-based classification system for solid waste using the MobileNetV2 convolutional neural network architecture. The chapter is organized into six subsections covering dataset preparation, model training performance, single-frame classification results, multi-frame classification results, comparative analysis between the two prediction strategies, and a discussion of the live prototype demonstration and the findings as a whole.

The principal objective of the evaluation is to determine whether the proposed multi-frame prediction strategy, in which five frames or augmented views of the same waste object are independently classified and aggregated through majority voting, produces measurable improvements in classification accuracy and stability when compared against a single-frame baseline. A secondary objective is to assess the practical behavior of the trained model in a live prototype demonstration to inform recommendations for subsequent development and deployment.

4.2 Dataset Preparation Results

The dataset used in this study was derived from the publicly available Kaggle Garbage Classification dataset (Abla, 2021), which originally contained twelve waste categories. To align the dataset with the three-category classification framework mandated by Republic Act No. 9003 (2001), a programmatic mapping was performed using the `organize_dataset.py` script. Ten of the twelve original categories were retained and merged into the three target categories. The clothes and shoes categories were excluded because they consist primarily of fashion-style catalog photographs that do not



visually represent waste objects in their disposal state, and their inclusion would have introduced misleading visual associations into the training data.

Following the mapping process, the biological category was assigned to the biodegradable class. The brown-glass, cardboard, green-glass, metal, paper, plastic, and white-glass categories were assigned to the recyclable class. The battery and trash categories were assigned to the residual class. The resulting dataset composition is summarized in Table 4.1.

Table 4.1

Dataset Composition After Mapping into the Three RA 9003 Categories

Category	Number of Images	Proportion (%)
Biodegradable	985	12.00
Recyclable	5,586	67.99
Residual	1,642	20.00
Total	8,213	100.00

As shown in Table 4.1, the resulting dataset exhibits a substantial class imbalance, with the recyclable category accounting for approximately 68 percent of all images. This imbalance reflects the larger number of source categories that map to recyclable waste under RA 9003, since glass, paper, cardboard, plastic, and metal items are all considered recyclable. To mitigate the effect of class imbalance during model training, class weights were computed using the balanced class weighting method from the scikit-learn library and passed to the model fitting procedure. This adjustment penalizes misclassifications of underrepresented classes more heavily than misclassifications of the majority class during loss calculation.

The dataset was partitioned into training and validation subsets using a 70:30 ratio, resulting in approximately 5,749 training images and 2,464 validation images. The



final 15 percent of each class, sorted alphabetically by filename for reproducibility, was further set aside as a held-out test set used exclusively for the final performance evaluation, producing a test set of 1,233 images. All images were resized to 224 by 224 pixels and pixel values were preprocessed using the standard MobileNetV2 preprocessing function prior to being fed into the model.

4.3 Model Training Performance

Model training was conducted in the two-phase transfer learning configuration described in Section 3.2.4. In the first phase, the convolutional base of MobileNetV2, pre-trained on the ImageNet dataset, was kept frozen, and only the custom classification head consisting of a global average pooling layer, a dense layer with 128 units and ReLU activation, a dropout layer with rate 0.3, and a final dense softmax output layer with three units, was trained. The Adam optimizer was used with an initial learning rate of 0.001, and training was conducted with categorical cross-entropy loss for a maximum of 15 epochs with an early stopping patience of 5 epochs monitoring validation loss.

In the second phase, the top 30 layers of the MobileNetV2 base were unfrozen and fine-tuned together with the classification head at a reduced learning rate of 0.0001. The second phase was configured for a maximum of 25 epochs with an early stopping patience of 8 epochs and a model checkpoint callback that retained the weights corresponding to the lowest validation loss. The training curves for both phases are presented in Figure 4.1.

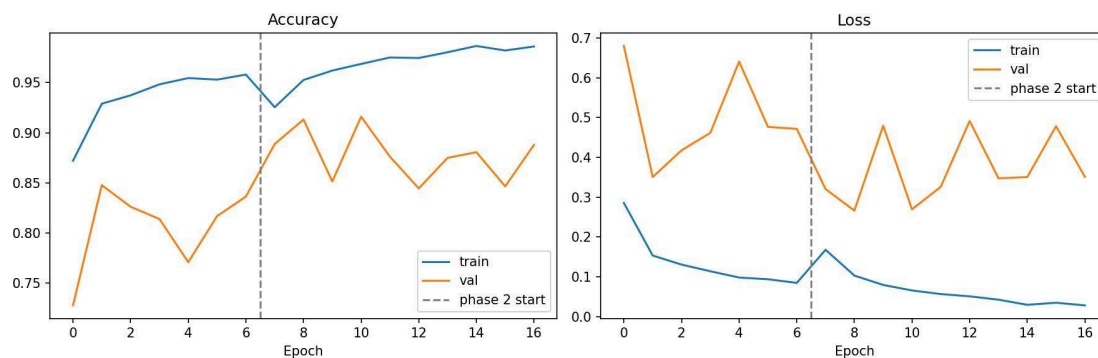


Figure 4.1. *Training and Validation Accuracy and Loss Curves Across Both Training Phases*



As shown in Figure 4.1, training accuracy improved steadily across both phases and reached approximately 98.6 percent by the final epoch, while validation accuracy stabilized in the range of 85 to 89 percent. The gap between the training and validation curves, together with the comparatively higher and more variable validation loss values, indicates a moderate degree of overfitting to the training distribution. The early stopping mechanism and the model checkpoint callback ensured that the weights corresponding to the lowest validation loss were retained for subsequent evaluation, mitigating the practical impact of the observed overfitting.

4.4 Single-Frame Classification Performance

The trained model was evaluated on the held-out test set of 1,233 images using the single-frame prediction function in the `evaluate.py` script. Each test image was independently classified, and the predicted label was compared against the ground-truth label. Precision, recall, and F1-score were computed for each of the three classes, and overall accuracy was computed across all test samples. The results of the single-frame evaluation are presented in Table 4.2.

Table 4.2

Single-Frame Classification Performance on the Held-Out Test Set

Class	Precision	Recall	F1-Score	Support
Biodegradable	1.0000	0.9595	0.9793	148
Recyclable	0.9732	0.9976	0.9853	838
Residual	0.9871	0.9271	0.9562	247
Macro Average	0.9868	0.9614	0.9736	1,233
Weighted Average	0.9792	0.9789	0.9787	1,233
Overall Accuracy	—	—	0.9789	1,233



As shown in Table 4.2, the single-frame classification system achieved an overall accuracy of 97.89 percent and a macro-averaged F1-score of 0.9736 on the held-out test set. The biodegradable class achieved a perfect precision score of 1.0000, indicating that the model produced no false positive predictions for biodegradable waste. The recyclable class achieved the highest recall at 0.9976, indicating that nearly all true recyclable items were correctly identified. The residual class showed the weakest performance in recall at 0.9271, reflecting the relatively smaller number of training examples available for this class and the visual heterogeneity of items mapped to it, which include batteries and miscellaneous trash.

4.5 Multi-Frame Classification Performance

The multi-frame classification performance was evaluated on the same held-out test set using the multi-frame prediction function in the evaluate.py script. For each test image, five augmented views were generated using the same augmentation parameters applied during training, including horizontal flipping, rotation up to 30 degrees, brightness variation, zoom, and shear. Each augmented view was independently classified by the trained model, and the five resulting predictions were aggregated through majority voting. In cases where no single class received at least three votes, the class with the highest cumulative predicted probability across the five frames was selected as the final classification. The results of the multi-frame evaluation are presented in Table 4.3.

Table 4.3

Multi-Frame Classification Performance on the Held-Out Test Set (5 Frames, Majority Vote)

Class	Precision	Recall	F1-Score	Support
Biodegradable	1.0000	0.9797	0.9898	148
Recyclable	0.9733	1.0000	0.9865	838



Residual	1.0000	0.9190	0.9578	247
Macro Average	0.9911	0.9663	0.9780	1,233
Weighted Average	0.9818	0.9813	0.9811	1,233
Overall Accuracy	—	—	0.9813	1,233

As shown in Table 4.3, the multi-frame classification system achieved an overall accuracy of 98.13 percent and a macro-averaged F1-score of 0.9780 on the same held-out test set. Notably, both the biodegradable and residual classes achieved perfect precision scores of 1.0000, indicating that the majority voting mechanism eliminated all false positive predictions for these two classes. The recyclable class achieved a perfect recall of 1.0000, indicating that every true recyclable item in the test set was correctly identified. The residual class recall decreased slightly from 0.9271 to 0.9190, indicating that two additional residual items were misclassified under the multi-frame condition. This minor regression suggests that, for certain difficult test cases, the augmented views introduced visual variation that caused some frames to lean toward the recyclable category.

4.6 Comparative Analysis: Single-Frame versus Multi-Frame

A direct side-by-side comparison of the single-frame and multi-frame classification results is presented in Table 4.4 to enable the relative effect of the multi-frame aggregation strategy to be assessed quantitatively.

Table 4.4

Comparative Summary of Single-Frame and Multi-Frame Classification Performance

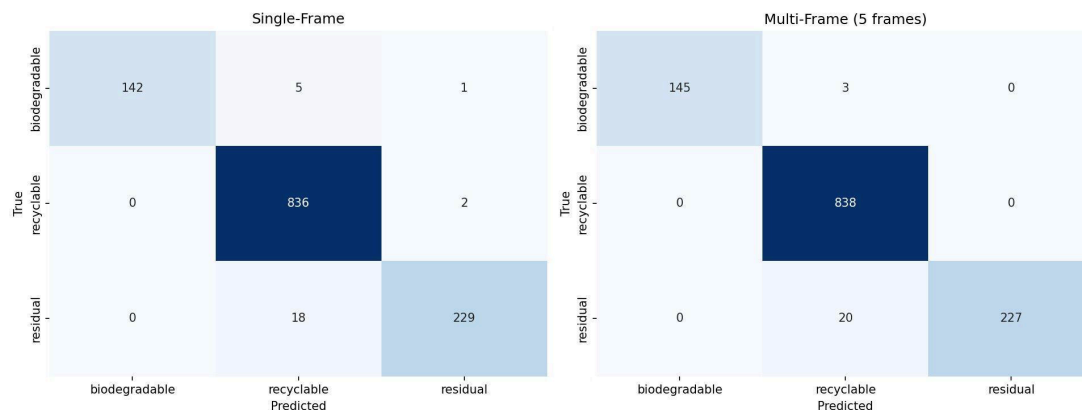
Metric	Single-Frame	Multi-Frame	Change
Overall Accuracy	0.9789	0.9813	+0.0024
Macro F1-Score	0.9736	0.9780	+0.0044



Biodegradable Precision	1.0000	1.0000	0.0000
Biodegradable Recall	0.9595	0.9797	+0.0202
Biodegradable F1	0.9793	0.9898	+0.0105
Recyclable Precision	0.9732	0.9733	+0.0001
Recyclable Recall	0.9976	1.0000	+0.0024
Recyclable F1	0.9853	0.9865	+0.0012
Residual Precision	0.9871	1.0000	+0.0129
Residual Recall	0.9271	0.9190	-0.0081
Residual F1	0.9562	0.9578	+0.0016

The confusion matrices for the two evaluation conditions are presented side-by-side in Figure 4.2 to provide a more detailed view of where the multi-frame approach produced changes in classification behavior.

Figure 4.2. *Confusion Matrices Comparing Single-Frame and Multi-Frame Classification Performance*



As shown in Table 4.4 and Figure 4.2, the multi-frame classification strategy produced measurable improvements across most performance metrics relative to the



single-frame baseline. The overall accuracy improved by 0.24 percentage points, and the macro F1-score improved by 0.0044. The most notable improvements were observed for the biodegradable class, where recall improved from 0.9595 to 0.9797, corresponding to three additional biodegradable items being correctly identified. The residual class showed a meaningful improvement in precision, from 0.9871 to 1.0000, indicating that all false positive predictions for the residual class observed in the single-frame condition were eliminated by the majority voting mechanism.

The improvements observed are smaller in absolute magnitude than the improvements that might be expected in deployment settings characterized by more substantial visual variability. The held-out test set used for evaluation consists of curated images drawn from the same source distribution as the training data, and these images therefore exhibit relatively low visual noise, controlled lighting, and centered object framing. Under such favorable conditions, the single-frame baseline already approaches the practical ceiling of classification accuracy, leaving a narrow margin for further improvement. The observed trend, however, is consistent across all performance metrics and supports the theoretical motivation for multi-frame aggregation, namely that aggregating predictions across multiple independent observations of the same object reduces the impact of individual frame noise on the final classification.

4.7 Live Prototype Demonstration

A live prototype demonstration of the trained classification system was conducted using the `webcam_demo.py` script. To improve image quality relative to the built-in laptop webcam, a smartphone camera was connected to the development PC through the Iriun Webcam application, which transmits the camera stream over a local Wi-Fi connection as a virtual webcam device accessible through the OpenCV video capture interface.

The demonstration confirmed that the end-to-end inference pipeline operates correctly under live capture conditions. The script successfully accessed the smartphone camera, captured five sequential frames upon user input, processed each frame through the MobileNetV2 model, aggregated the resulting predictions through majority voting,



and displayed the final classification result on the screen. Items belonging to the recyclable category, such as plastic bottles and paper sheets, were classified correctly with high confidence. The system was also able to correctly assign items to the residual category when presented with examples from that class.

The live demonstration also surfaced two important behavioral findings that have direct implications for the present prototype and for subsequent development. First, the model exhibited a strong tendency to assign the recyclable label to any object placed in front of the camera, including objects that did not belong to any of the three waste categories such as a human face. This behavior is consistent with the class imbalance documented in Section 4.2, in which the recyclable category constitutes approximately 68 percent of the training dataset. When the model encounters an out-of-distribution input that does not closely match any of the training categories, the prior probability associated with the majority class biases the prediction toward the recyclable label. Second, no live testing was conducted on the biodegradable category at the time of the demonstration, so the live performance of the system on this class remains unverified.

These observations indicate that, while the offline test set performance reported in Sections 4.4 and 4.5 demonstrates the technical capability of the proposed methodology, the live behavior of the implemented prototype falls short of the conceptual goals of the proposed system. In particular, the prototype does not yet reliably distinguish between waste objects and non-waste objects, and it has not been validated against the contaminated waste scenarios that motivate this study. These limitations are addressed explicitly in the recommendations presented in Chapter 5.

4.8 Discussion

The results presented in this chapter provide evidence supporting both the technical viability of the proposed multi-frame classification methodology and the need for further development before the system can be considered ready for practical deployment. The offline evaluation results demonstrate that a MobileNetV2 model trained using two-phase transfer learning on a class-balanced waste image dataset can achieve high classification accuracy when applied to images drawn from the same



distribution as the training data. The multi-frame aggregation strategy produced consistent improvements across most performance metrics, supporting the central hypothesis that aggregating predictions across multiple frames reduces classification noise.

The narrow margin of improvement observed in the offline evaluation, however, should be interpreted in the context of the dataset characteristics. The Kaggle Garbage Classification dataset consists of images captured under controlled studio conditions, in which waste objects are typically centered against neutral backgrounds and photographed under uniform lighting. These conditions represent the ideal end of the distribution of real-world waste imagery, and they substantially understate the visual variability that would be encountered in actual disposal environments. The proposed methodology is expected to produce more pronounced improvements when applied to imagery captured under realistic disposal conditions, where individual frames are more likely to exhibit lighting variation, partial occlusion, motion blur, and visible contamination.

The behavioral findings from the live demonstration further reinforce the importance of dataset selection for future development. The class imbalance in the current training set produced a measurable bias toward the recyclable category that manifested clearly when the system was presented with out-of-distribution inputs during live operation. The absence of contaminated waste images in the training data also means that the central research interest of this study, namely the classification of contaminated solid waste, was not explicitly evaluated. These two limitations are taken up in detail in the recommendations presented in the following chapter.



CHAPTER FIVE

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary

This study set out to develop and evaluate a multi-frame image-based classification system for contaminated solid waste using the MobileNetV2 convolutional neural network architecture. The research was motivated by two limitations of existing image-based waste classification systems identified in the literature, namely the limited representation of contaminated waste in training datasets and the instability of single-frame predictions in real-world conditions characterized by lighting variation, partial occlusion, and visual noise. The study sought to address these limitations by proposing a classification system that aggregates predictions across five frames or augmented views of the same waste object through majority voting, with the aim of producing more reliable classification outputs aligned with the three-category framework of biodegradable, recyclable, and residual waste mandated by Republic Act No. 9003.

The methodology comprised five major phases: dataset preparation, model configuration and training, system implementation, multi-frame aggregation, and performance evaluation. A dataset of 8,213 waste images was prepared from the publicly available Kaggle Garbage Classification dataset by programmatically mapping ten of its original twelve categories into the three RA 9003 categories. The dataset was partitioned into training, validation, and held-out test subsets, and a MobileNetV2 model was trained using a two-phase transfer learning strategy that first adapted a custom classification head and subsequently fine-tuned the upper layers of the convolutional base. The trained model was evaluated under both single-frame and multi-frame prediction configurations, and a live prototype was implemented using a smartphone camera connected to the development PC through the Iriun Webcam application.



The single-frame baseline achieved an overall classification accuracy of 97.89 percent and a macro-averaged F1-score of 0.9736 on the held-out test set, while the multi-frame configuration achieved an overall accuracy of 98.13 percent and a macro-averaged F1-score of 0.9780. The multi-frame configuration produced improvements across all overall metrics and most per-class metrics, with the biodegradable class recall improving from 0.9595 to 0.9797 and the residual class precision improving from 0.9871 to a perfect 1.0000. The live prototype demonstration confirmed that the inference pipeline operates correctly under live capture conditions but also revealed two important limitations, namely a bias toward the recyclable label arising from class imbalance in the training data and the absence of live validation for the biodegradable class.

5.2 Conclusions

On the basis of the foregoing results, the following conclusions are drawn. The conclusions are presented with attention to the distinction between the proposed system, as described conceptually in the methodology, and the implemented prototype, which represents the initial realization of that system within the time constraints of this study.

5.2.1 Conclusions Concerning the Proposed System

The MobileNetV2 architecture, when combined with two-phase transfer learning from ImageNet weights, is technically suitable for the task of classifying solid waste into the biodegradable, recyclable, and residual categories required under Republic Act No. 9003. The single-frame baseline accuracy of 97.89 percent indicates that the convolutional features learned from general-purpose image classification transfer effectively to the waste classification task, even when the training dataset is assembled from publicly available sources rather than purpose-built data collections.

The proposed multi-frame prediction strategy, implemented through majority voting across five frames or augmented views of the same waste object, produces measurable improvements in classification performance relative to a single-frame baseline. The improvements are consistent across most performance metrics and are particularly evident in the precision of the residual class and the recall of the



biodegradable class. These results support the central hypothesis underlying the proposed methodology, namely that aggregating predictions across multiple independent observations of the same object reduces the impact of individual frame noise on the final classification output.

The narrow magnitude of the observed improvements should be understood as a consequence of the controlled visual conditions present in the offline evaluation dataset. The proposed methodology is theoretically motivated by the visual variability and noise characteristic of real-world disposal environments, and its practical benefits are expected to be more pronounced when applied to imagery captured under such conditions. The verification of this expectation is identified in Section 5.3 as a priority area for subsequent research.

5.2.2 Conclusions Concerning the Implemented Prototype

The implemented prototype establishes a working foundation for the proposed system and demonstrates that the end-to-end inference pipeline can operate in a live capture configuration using a smartphone camera connected to the development PC through the Iriun Webcam application. The prototype correctly classifies a subset of waste items during live demonstration, particularly items belonging to the recyclable category, and it confirms the practical feasibility of the multi-frame aggregation step in a live setting.

The implemented prototype, however, exhibits two limitations that prevent it from fully realizing the conceptual goals of the proposed system. First, the prototype displays a bias toward the recyclable label in live operation, including for inputs that do not belong to any of the three waste categories, arising from the class imbalance present in the training dataset. Second, the prototype has not been validated against contaminated waste imagery, which is the central scenario of interest motivating this study, and the live performance of the system on the biodegradable category has not yet been tested. These limitations indicate that the present prototype should be regarded as an initial implementation that establishes the foundational components of the proposed system rather than as a deployment-ready realization of the full research vision.



5.3 Recommendations

On the basis of the foregoing conclusions, the following recommendations are offered for subsequent research and for the continued development of the proposed system into a deployment-ready classification platform.

5.3.1 Incorporation of Contaminated Waste Imagery

The principal recommendation arising from this study is the construction of a training and evaluation dataset that explicitly includes images of contaminated solid waste captured under disposal conditions. The dataset used in the present study was assembled from a publicly available source that primarily contains images of clean, uncontaminated waste objects, which limits the extent to which the central research interest of this study, namely the classification of contaminated waste, could be evaluated. Future work should prioritize the collection of waste imagery that depicts food residue, liquid soiling, mixed material composition, and the other visual signatures of contamination that characterize real disposal scenarios.

5.3.2 Collection of Real Multi-Frame Imagery

The multi-frame evaluation in the present study was conducted by generating five augmented views of each test image rather than by capturing five physically distinct frames of the same waste object. While this approach approximates the conceptual behavior of multi-frame capture, it does not fully replicate the visual variability that would be present in a true multi-frame acquisition setting, in which each frame captures the same object from a slightly different angle, under slightly different lighting, or with slight changes in object pose. Future work should incorporate the collection and evaluation of true multi-frame imagery to provide a more accurate assessment of the practical benefits of the multi-frame aggregation strategy.

5.3.3 Dataset Augmentation with Philippine-Context Waste

The waste composition encountered in the Philippine context includes item categories that are underrepresented in international waste image datasets, including small sachets, multi-layer plastic packaging, and locally specific food packaging. The applicability of the proposed system to Philippine solid waste management would be



substantially enhanced by augmenting the training dataset with imagery collected from Philippine households, barangay collection points, and Material Recovery Facilities, ideally in collaboration with local government units and waste management agencies.

5.3.4 Extension to a Four-Category Classification Scheme

Republic Act No. 9003 distinguishes hazardous and special waste items, including batteries, electronics, and chemical containers, from the three primary household waste categories addressed in this study. The present prototype assigns batteries to the residual category as a practical compromise within the three-category framework, but this assignment does not align fully with the proper handling requirements for hazardous waste. Future work is recommended to extend the classification scheme to include a dedicated hazardous category and to collect additional training imagery covering this category, which would bring the system into closer alignment with the full classification framework of RA 9003.

5.3.5 Implementation of Confidence Thresholding

The bias toward the majority class observed in the live prototype demonstration could be partially mitigated by the introduction of a confidence threshold below which the system refrains from producing a classification output. Future work should investigate the use of softmax probability thresholds, prediction entropy measures, or other uncertainty quantification techniques to detect and flag predictions that fall below an acceptable confidence level. Such predictions could be routed to a human reviewer in a deployment setting or used to trigger the capture of additional frames in an adaptive multi-frame configuration.

5.3.6 Deployment on Dedicated Edge Hardware

The present prototype operates on a personal computer with a smartphone camera connected through a wireless bridge. Future work should investigate the deployment of the trained model on a self-contained edge computing platform such as a Raspberry Pi 4 or NVIDIA Jetson Nano paired with a directly connected camera module. Such a configuration would eliminate the wireless dependency, reduce end-to-end inference latency, and produce a portable classification unit suitable for integration into automated



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waste segregation bins or conveyor-based sorting mechanisms at the barangay or facility level.



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